

train__presence__classifier

February 5, 2026

1 Embryo Presence Classifier

1.1 Annotation Dictionary for Embryo Presence Classification

Each entry in the dictionary represents one embryo sample, and the associated list of tuples defines inclusive frame index ranges (start, end) where the embryo is absent.

```
[1]: NO_EMBRYO_RANGES = {
    "AAL839-6" : [(475, 558)],
    "AB91-1" : [(376, 558)],
    "AC264-1" : [(209, 343)],
    "ADM715-1-2" : [(358, 446)],
    "AG782-8" : [(375, 565)],
    "AH988-4" : [(471, 551)],
    "AHS115-5" : [(272, 293), (474, 557)],
    "AHS599-4" : [(459, 464)],
    #"ALR493-6" : [(470, 566)], Truncated images
    "AM33-2" : [(287, 562)],
    "AM716-1" : [(476, 572)],
    "AM716-7" : [(476, 572)],
    "AMT360-1-9" : [(390, 394)],
    "AS556-3" : [(473, 560)],
    "AS662-2" : [(275, 466)],
    "AS1015-2" : [(297, 371)],
    "BA518-7" : [(464, 553)],
    "BA560-1" : [(351, 380)], # Darkens in this range
    "BA560-2" : [(476, 574)],
    "BA782-2" : [(1, 1), (373, 376)],
    "BA1195-9" : [(256, 276), (377, 663)],
    "BC254-5" : [(381, 561)],
    "BC277-9" : [(366, 458)],
    "BC277-10" : [(179, 264)], # Darkens in this range
    "BC518-6" : [(419, 551)],
    "BC750-7" : [(465, 555)],
    "BC973-4" : [(473, 555)],
    "BE327-2" : [(168, 283), (473, 574)],
    "BE413-3" : [(353, 418)],
    "BE645-7" : [(375, 559)],
```

```

"BF924-3" : [(471, 557)],
"BG201-1" : [(191, 274)],
"BG723-5" : [(377, 555)],
"BH034-4" : [(374, 564)],
"BI224-2" : [(357, 423)],
#"BJ492-8" : [(145, 147)], Truncated Images
"BJ497-7" : [(190, 460)],
"BJ3371-9" : [(275, 413)],
"BL042-8" : [(478, 564)],
"BL285-1-3" : [(473, 567)],
"BL366-6" : [(470, 559)],
"BL526-1" : [(372, 656)],
"BL526-5" : [(1, 13), (468, 656)],
"BM007-9" : [(375, 560)],
"BM076-4" : [(466, 554)],
"BM201-1" : [(374, 569)],
"BM256-1" : [(460, 555)],
"BM352-6" : [(377, 567)],
"BM414-3" : [(350, 413)],
"BM968-3" : [(295, 362)],
"BMH857-2" : [(367, 456)],
"BMS819-7" : [(484, 569)],
"BN356-3" : [(357, 423)],
"BO613-7" : [(462, 564)],
"BR953-7" : [(468, 563)],
"BS495-9" : [(472, 560)],
"BS648-7" : [(449, 516)],
"BS777-8" : [(404, 755)],
"BS836-11" : [(453, 545)],
"BS918-6" : [(558, 562)],
"BS1033-2" : [(370, 557)],
"BS1086-1" : [(480, 570)],
"BY829-3" : [(360, 453)],
"CA063-6" : [(200, 276), (464, 552)],
"CA364-7" : [(365, 552)],
}

EXCLUDE_FROM_TRAIN = {"AM365-7"} # blurred embryo - keep for later tests
TRAIN_EMBRYOS = sorted(NO_EMBRYO_RANGES.keys())

```

1.2 Setup: Imports, Configuration, and Utilities

```

[2]: # Standard libs
import os, math, random, warnings
from pathlib import Path
from collections import Counter

```

```

# Numerics / data
import numpy as np
import pandas as pd

# Imaging & plots
from PIL import Image
import matplotlib.pyplot as plt

# Notebook display
from IPython.display import display

# Matplotlib defaults
plt.rcParams['figure.figsize'] = (10, 10)
plt.rcParams['axes.grid'] = False
warnings.filterwarnings("ignore")

# Reproducibility
RNG_SEED = 42
random.seed(RNG_SEED)
np.random.seed(RNG_SEED)

# RegEx support in python
import re

def extract_index(fp):
    """
    Searches for a pattern like "RUN12345" and extracts the numeric part.
    If the pattern isn't found, it returns None.
    Input:
        fp: A file-like object with a 'name' attribute (e.g., a Path object).
    Output:
        An integer representing the extracted index, or None if not found.
    """
    m = re.search(r"RUN(\d+)", fp.name)
    return int(m.group(1)) if m else None

```

1.2.1 Path Configuration and Data Validation

Establishes the directory structure and validates that required data directories exist before the training pipeline proceeds.

```

[3]: # a Path object pointing to the parent directory of the current notebook
      ↪ location,
      # then calls .resolve() to convert it into an absolute filesystem path.
      PROJECT_ROOT = Path('..').resolve()

      # Directories for images and annotations

```

```

IMG_DIR = PROJECT_ROOT / 'data' / 'embryo_dataset_bronze'
ANN_DIR = PROJECT_ROOT / 'data' / 'embryo_dataset_annotations'

# Assertions to ensure directories exist
assert IMG_DIR.exists(), f"Not found: {IMG_DIR} <-- check your folder
↳structure"
assert ANN_DIR.exists(), f"Not found: {ANN_DIR} <-- check your folder
↳structure"

# Focal planes to train on
FOCAL_PLANES = ['F0', 'F-15', 'F-30', 'F-45', 'F15', 'F30', 'F45']

print(f"Focal planes: {FOCAL_PLANES}")
print(f>Data root: {IMG_DIR}")

```

Focal planes: ['F0', 'F-15', 'F-30', 'F-45', 'F15', 'F30', 'F45']

Data root:

/mnt/c/Users/ioann/OneDrive/Documents/Projects/thesis/data/embryo_dataset_bronze

```

[4]: from torch.utils.data import Dataset, DataLoader
import torch, torch.nn as nn

# Dataset for embryo presence classification
class EmbryoPresenceDataset(Dataset):
    """
    Dataset for embryo presence classification
    """
    def __init__(self, img_dir, embryo_ids, focal_planes, no_embryo_ranges,
↳transform=None):
        """
        Initializes the dataset with image directory, embryo IDs, focal planes,
↳and ranges without embryos.
        Inputs:
            img_dir: Path to the root image directory.
            embryo_ids: List of embryo IDs to include.
            focal_planes: List of focal planes to include.
            no_embryo_ranges: Dict mapping embryo IDs to lists of (start, end)
↳tuples indicating frame ranges without embryos.
            transform: Optional transform to apply to images.
        """
        self.samples = [] # list of (path, label, focal_plane)
        self.transform = transform
        skipped_count = 0 # count of corrupted frames skipped

        # Iterate over focal planes
        for fp in focal_planes:
            fp_dir = img_dir / fp # e.g., IMG_DIR / 'F0'

```

```

if not fp_dir.is_dir():
    print(f" focal plane {fp_dir} missing - skipped")
    continue
# Iterate over embryo IDs
for eid in embryo_ids:
    folder = fp_dir / eid # e.g., IMG_DIR / 'F0' / 'AAL839-6'
    if not folder.is_dir():
        continue

    # collect frame paths with index
    idx2path = {extract_index(fpath): fpath
                 for fpath in folder.glob("*.jpg")}
    if extract_index(fpath) is not None}

    # enumerate frames and filter corrupted ones at build time
    for idx, fpath in idx2path.items():
        # try to open and verify image is readable
        try:
            with Image.open(fpath) as im:
                im.convert("L") # verify grayscale conversion works
        except (OSError, ValueError) as e:
            skipped_count += 1
            continue

        # decide label based on no_embryo_ranges
        absent = any(s <= idx <= e for (s, e) in
↪no_embryo_ranges[eid])
        label = 0 if absent else 1
        self.samples.append((fpath, label, fp))

    print(f"Dataset built with {len(self.samples):,} frames across
↪{len(focal_planes)} focal planes (skipped {skipped_count} corrupted)")

def __len__(self):
    """
    Returns the total number of samples in the dataset.
    :return: Integer count of samples.
    """
    return len(self.samples)

def __getitem__(self, i):
    path, label, fp = self.samples[i]
    with Image.open(path) as im:
        im = im.convert("L") # 1-channel grayscale
        if self.transform: im = self.transform(im)
    return im, torch.tensor(label, dtype=torch.long)

```

1.2.2 Transforms & Dataloader

```
[5]: import torchvision.transforms as T
import torchvision.transforms.v2 as v2
from torch.utils.data import WeightedRandomSampler

IMG_SIZE = 224

# Base transforms: resize, convert to tensor, normalize
tfm = T.Compose([
    T.Resize((IMG_SIZE, IMG_SIZE)),
    T.ToTensor(),
    T.Normalize([0.5], [0.5]), # normalize to [-1, 1]
])

# Train augmentations applied BEFORE ToTensor & Normalize
train_tfm = T.Compose([
    T.Resize((IMG_SIZE, IMG_SIZE)),
    T.RandomHorizontalFlip(p=0.5),
    T.RandomVerticalFlip(p=0.5),
    T.RandomRotation(45),
    T.RandomAffine(degrees=0, scale=(0.8, 1.0)), # random scale
    T.GaussianBlur(kernel_size=3, sigma=(0.1, 2.0)),
    T.ToTensor(),
    T.Normalize([0.5], [0.5]), # normalize to [-1, 1]
])

# --- embryo-level train / val / test split -----
import random
random.seed(42)

all_embryos = TRAIN_EMBRYOS.copy()
random.shuffle(all_embryos)

val_frac = 0.20 # 10% of embryos
test_frac = 0.20 # 10% of embryos

n_val = int(len(all_embryos) * val_frac)
n_test = int(len(all_embryos) * test_frac)

val_embryos = all_embryos[:n_val]
test_embryos = all_embryos[n_val:n_val + n_test]
train_embryos = all_embryos[n_val + n_test:]

print(f"Train embryos: {len(train_embryos)} | Val embryos: {len(val_embryos)} |
↳ Test embryos: {len(test_embryos)}")
```

```

# Build three **separate** datasets so no embryo appears in two splits
train_ds = EmbryoPresenceDataset(IMG_DIR, train_embryos, FOCAL_PLANES,
    ↪NO_EMBRYO_RANGES, transform=train_tfm)
val_ds   = EmbryoPresenceDataset(IMG_DIR, val_embryos,   FOCAL_PLANES,
    ↪NO_EMBRYO_RANGES, transform=tfm)
test_ds  = EmbryoPresenceDataset(IMG_DIR, test_embryos,  FOCAL_PLANES,
    ↪NO_EMBRYO_RANGES, transform=tfm)

# Class weights for imbalance: weight inversely by class frequency
train_labels = [lbl for _, lbl, _ in train_ds.samples]
class_counts = np.bincount(train_labels)
class_weights = 1.0 / (class_counts + 1e-6) # avoid division by zero
class_weights = class_weights / class_weights.sum() * len(class_weights) #
    ↪normalize
sample_weights = np.array([class_weights[lbl] for lbl in train_labels])

weighted_sampler = WeightedRandomSampler(weights=sample_weights,
    ↪num_samples=len(train_ds), replacement=True)

BATCH_SIZE = 64
train_dl = DataLoader(train_ds, batch_size=BATCH_SIZE,
    ↪sampler=weighted_sampler, num_workers=8, pin_memory=True)
val_dl   = DataLoader(val_ds,   batch_size=BATCH_SIZE, shuffle=False,
    ↪num_workers=8, pin_memory=True)
test_dl  = DataLoader(test_ds,  batch_size=BATCH_SIZE, shuffle=False,
    ↪num_workers=8, pin_memory=True)

print(f"Class distribution in train set: {dict(zip(['NO_EMBRYO', 'EMBRYO'],
    ↪class_counts))}")
print(f"Class weights applied: {dict(zip(['NO_EMBRYO', 'EMBRYO'],
    ↪class_weights))}")
# -----

```

Train embryos: 39 | Val embryos: 12 | Test embryos: 12

```

-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[5], line 46

```

```

    43 print(f"Train embryos: {len(train_embryos)} | Val embryos:
    ↪{len(val_embryos)} | Test embryos: {len(test_embryos)}")
    45 # Build three **separate** datasets so no embryo appears in two splits
---> 46 train_ds =
    ↪EmbryoPresenceDataset(IMG_DIR, train_embryos, FOCAL_PLANES, NO_EMBRYO_RANGES, transform=tr
    47 val_ds   = EmbryoPresenceDataset(IMG_DIR, val_embryos,   FOCAL_PLANES,
    ↪NO_EMBRYO_RANGES, transform=tfm)
    48 test_ds  = EmbryoPresenceDataset(IMG_DIR, test_embryos,  FOCAL_PLANES,
    ↪NO_EMBRYO_RANGES, transform=tfm)

```

```

Cell In[4], line 45, in EmbryoPresenceDataset.__init__(self, img_dir,
↳ embryo_ids, focal_planes, no_embryo_ranges, transform)
    43 try:
    44     with Image.open(fpath) as im:
--> 45         im.convert("L") # verify grayscale conversion works
    46 except (OSError, ValueError) as e:
    47     skipped_count += 1

```

```

File ~/miniconda3/envs/thesis/lib/python3.10/site-packages/PIL/Image.py:967, in
↳ Image.convert(self, mode, matrix, dither, palette, colors)
    915 def convert(
    916     self,
    917     mode: str | None = None,
    (...)
    921     colors: int = 256,
    922 ) -> Image:
    923     """
    924     Returns a converted copy of this image. For the "P" mode, this
    925     method translates pixels through the palette. If mode is
    (...)
    964     :returns: An :py:class:`~PIL.Image.Image` object.
    965     """
--> 967     self.load()
    969     has_transparency = "transparency" in self.info
    970     if not mode and self.mode == "P":
    971         # determine default mode

```

```

File ~/miniconda3/envs/thesis/lib/python3.10/site-packages/PIL/ImageFile.py:386
↳ in ImageFile.load(self)
    384         read_bytes = next_offset - offset
    385 try:
--> 386     s = read(read_bytes)
    387 except (IndexError, struct.error) as e:
    388     # truncated png/gif
    389     if LOAD_TRUNCATED_IMAGES:

```

```

File ~/miniconda3/envs/thesis/lib/python3.10/site-packages/PIL/JpegImagePlugin.
↳ py:418, in JpegImageFile.load_read(self, read_bytes)
    412 """
    413 internal: read more image data
    414 For premature EOF and LOAD_TRUNCATED_IMAGES adds EOI marker
    415 so libjpeg can finish decoding
    416 """
    417 assert self.fp is not None
--> 418 s = self.fp.read(read_bytes)
    420 if not s and ImageFile.LOAD_TRUNCATED_IMAGES and not hasattr(self,
↳ "_ended"):

```



```

421     # Premature EOF.
422     # Pretend file is finished adding EOF marker
423     self._ended = True

```

KeyboardInterrupt:

1.2.3 Model Definition

```

[ ]: import torchvision.models as models

device = "cuda" if torch.cuda.is_available() else "cpu"

# Build model: ResNet18 adapted for 1-channel input and 2 output classes
model = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)

# Adapt the first convolutional layer to accept 1-channel input instead of 3
model.conv1 = nn.Conv2d(1, 64, kernel_size=7, stride=2, padding=3, bias=False)

# Modify the final fully connected layer for 2 classes
model.fc = nn.Linear(model.fc.in_features, 2) # 2 classes
model.to(device)

criterion = nn.CrossEntropyLoss() # no weight param; using
↳ WeightedRandomSampler instead
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4)

```

1.2.4 Training Loop

```

[ ]: import copy
from tqdm.auto import tqdm

EPOCHS = 50
best_val_loss = float("inf")
patience = 10 # epochs without improvement
epochs_no_improve = 0
best_model_wts = copy.deepcopy(model.state_dict())

for epoch in range(1, EPOCHS+1):
    # ---- train ----
    model.train()
    running_loss, correct = 0., 0
    for x, y in tqdm(train_dl, desc=f"Epoch {epoch}", leave=False):
        # Augmentations already applied in dataset via train_tfm; no additional
        ↳ augmentation needed
        x, y = x.to(device), y.to(device)

```

```

optimizer.zero_grad()
y_hat = model(x)
loss = criterion(y_hat, y)
loss.backward()
optimizer.step()

running_loss += loss.item() * x.size(0)
correct += (y_hat.argmax(1) == y).sum().item()

train_loss = running_loss / len(train_ds)
train_acc = correct / len(train_ds)

# ---- validate ----
model.eval()
running_vloss, val_correct = 0.0, 0
with torch.no_grad():
    for x, y in val_dl:
        x, y = x.to(device), y.to(device)
        y_hat = model(x)
        vloss = criterion(y_hat, y)
        running_vloss += vloss.item() * x.size(0)
        val_correct += (y_hat.argmax(1) == y).sum().item()

val_loss = running_vloss / len(val_ds)
val_acc = val_correct / len(val_ds)
# ----- early-stopping on val_loss -----
if val_loss < best_val_loss - 1e-6: # small delta to avoid ties
    best_val_loss = val_loss
    epochs_no_improve = 0
    best_model_wts = copy.deepcopy(model.state_dict())
else:
    epochs_no_improve += 1
    if epochs_no_improve >= patience:
        print(f"No val-loss improvement for {patience} epochs → stopping_
↪early.")
        break

    print(f"[{epoch}] loss {train_loss:.4f} | train_acc {train_acc:.3f} | val_
↪loss {val_loss:.4f} | val acc {val_acc:.3f}")

model.load_state_dict(best_model_wts)

```

Epoch 1: 0% | 0/2191 [00:00<?, ?it/s]

[1] loss 0.0052 | train_acc 0.998 | val loss 0.0390 | val acc 0.993

Epoch 2: 0% | 0/2191 [00:00<?, ?it/s]

```

[2] loss 0.0028 | train_acc 0.999 | val loss 0.0332 | val acc 0.993
Epoch 3:   0%|          | 0/2191 [00:00<?, ?it/s]
[3] loss 0.0026 | train_acc 0.999 | val loss 0.0531 | val acc 0.994
Epoch 4:   0%|          | 0/2191 [00:00<?, ?it/s]
[4] loss 0.0023 | train_acc 0.999 | val loss 0.0439 | val acc 0.996
Epoch 5:   0%|          | 0/2191 [00:00<?, ?it/s]

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0><function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process      assert self._parent_pid ==
os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0><function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```

```

packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/mult
iprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: assert self._parent_pid == os.getpid(), 'can only test a
child process'
AssertionError: can only test a child processcan only test a child process

Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: self._shutdown_workers()
Exception ignored in: self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
if w.is_alive():
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive
self._shutdown_workers()
assert self._parent_pid == os.getpid(), 'can only test a child process'
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
self._shutdown_workers()
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
: can only test a child process
if w.is_alive():AssertionError File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
: can only test a child process
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionErrorassert self._parent_pid == os.getpid(), 'can only test a child
process'
AssertionError: can only test a child process: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):

```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1604, in __del__
```

```
Traceback (most recent call last):
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1604, in __del__
```

```
    self._shutdown_workers()
```

```
self._shutdown_workers()
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
```

```
    if w.is_alive():Exception ignored in: File
```

```
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
```

```
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
```

```
    if w.is_alive():Exception ignored in:
```

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
```

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0> File "/home/  
/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line  
160, in is_alive
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/  
process.py", line 160, in is_alive
```

```
assert self._parent_pid == os.getpid(), 'can only test a child process'Traceback  
(most recent call last):
```

```
assert self._parent_pid == os.getpid(), 'can only test a child process'Traceback  
(most recent call last):
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1604, in __del__
```

```
AssertionError File
```

```
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
```

```
packages/torch/utils/data/dataloader.py", line 1604, in __del__
```

```
AssertionError : can only test a child processself._shutdown_workers(): can  
only test a child processself._shutdown_workers()
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-  
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
```

```
    if w.is_alive():
```

```
if w.is_alive():
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc  
ess.py", line 160, in is_alive
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/  
process.py", line 160, in is_alive
```

```

    assert self._parent_pid == os.getpid(), 'can only test a child
process'Exception ignored in:
assert self._parent_pid == os.getpid(), 'can only test a child process'Exception
ignored in:
<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>AssertionError<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>AssertionError
: Traceback (most recent call last):
can only test a child process
: Traceback (most recent call last):
can only test a child process  File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

    self._shutdown_workers()self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():    if w.is_alive():
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

```

```

        assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError    assert self._parent_pid == os.getpid(), 'can only test a child
process'

```

```

AssertionError: can only test a child process

```

```

: can only test a child process

```

```

Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

    self._shutdown_workers()

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

    self._shutdown_workers()

```

```

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```

```

packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):

Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```



```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError:
AssertionError: can only test a child process
can only test a child process
Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multip
rocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: : can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child processAssertionError: can only test a
child process

```

```

Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/pro
cess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
Exception ignored in:      assert self._parent_pid == os.getpid(), 'can only test
a child process'
AssertionError<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
Exception ignored in:      assert self._parent_pid == os.getpid(), 'can only test
a child process'
AssertionError<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
: AssertionErrorcan only test a child process

```

```

: : AssertionErrorcan only test a child process
: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process':
AssertionErrorif w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process':
AssertionErrorcan only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc

```

```

ess.py", line 160, in is_alive
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: assert self._parent_pid == os.getpid(), 'can only test a child
process'
AssertionError: can only test a child process
: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
if w.is_alive():
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():

```

```

self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```

```

packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError:      if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multip
rocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
: can only test a child process
Exception ignored in: Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

        self._shutdown_workers()
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

[5] loss 0.0019 | train_acc 1.000 | val loss 0.0386 | val acc 0.994
Epoch 6:   0%|          | 0/2191 [00:00<?, ?it/s]
[6] loss 0.0018 | train_acc 0.999 | val loss 0.0442 | val acc 0.994
Epoch 7:   0%|          | 0/2191 [00:00<?, ?it/s]
[7] loss 0.0015 | train_acc 1.000 | val loss 0.0579 | val acc 0.994
Epoch 8:   0%|          | 0/2191 [00:00<?, ?it/s]
[8] loss 0.0018 | train_acc 0.999 | val loss 0.0428 | val acc 0.995
Epoch 9:   0%|          | 0/2191 [00:00<?, ?it/s]

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():

```



```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError:
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```

```

packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
        self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child
process'self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'if
w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: can only test a child process

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

```

ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

    if w.is_alive():
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
            assert self._parent_pid == os.getpid(), 'can only test a child process'
        AssertionError: can only test a child process
        assert self._parent_pid == os.getpid(), 'can only test a child process'
        AssertionError: can only test a child process
        Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
    Traceback (most recent call last):
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
        Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
    Traceback (most recent call last):
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
            self._shutdown_workers()
            self._shutdown_workers()
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
            if w.is_alive():
                File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
                    assert self._parent_pid == os.getpid(), 'can only test a child process'
                AssertionError: can only test a child process
                assert self._parent_pid == os.getpid(), 'can only test a child process'
                AssertionError: can only test a child process
                Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
            Traceback (most recent call last):
                File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
                Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
        Traceback (most recent call last):
            File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
                self._shutdown_workers()
            File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
                if w.is_alive():
                    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
                        assert self._parent_pid == os.getpid(), 'can only test a child process'
                    
```

```

ess.py", line 160, in is_alive
self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process    assert self._parent_pid ==
os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

    if w.is_alive():Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

```

```

    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```

```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):

```

```

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```

```

packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    Traceback (most recent call last):
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
    AssertionError: can only test a child processif w.is_alive():
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
    AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()

```



```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0> File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0> File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

        assert self._parent_pid == os.getpid(), 'can only test a child
process'
Traceback (most recent call last):

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    AssertionError if w.is_alive():
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

        assert self._parent_pid == os.getpid(), 'can only test a child
process'
Traceback (most recent call last):

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    AssertionError: can only test a child processself._shutdown_workers()
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

        : can only test a child processself._shutdown_workers()
      File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

        if w.is_alive():
          File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
Exception ignored in: assert self._parent_pid == os.getpid(), 'can only test
a child process'
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0> if
w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
Exception ignored in: assert self._parent_pid == os.getpid(), 'can only test

```

```
a child process'
<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>AssertionError: can only test a child process
```

```
Traceback (most recent call last):
AssertionError: can only test a child process
```

```
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in:      File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in:      <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
if w.is_alive():
```

```
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
if w.is_alive():
```

```
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
assert self._parent_pid == os.getpid(), 'can
only test a child process'
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
AssertionError: self._shutdown_workers()
assert self._parent_pid ==
os.getpid(), 'can only test a child process'
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
AssertionError: can only test a child process
if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
```

```

ess.py", line 160, in is_alive
: can only test a child process
if w.is_alive():
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in:      assert self._parent_pid == os.getpid(), 'can only test
a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():<function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        self._shutdown_workers()      Exception ignored in:
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        assert self._parent_pid == os.getpid(), 'can only test a child process'<function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

AssertionError      : Traceback (most recent call last):
can only test a child processif w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
self._shutdown_workers()      Exception ignored in:
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
assert self._parent_pid == os.getpid(), 'can only test a child process'<function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

AssertionError      : Traceback (most recent call last):
can only test a child processif w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

```

```

self._shutdown_workers() File "/home/khasion/miniconda3/envs/thesis/lib/python3
.10/multiprocessing/process.py", line 160, in is_alive

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

```

```

self._shutdown_workers() File "/home/khasion/miniconda3/envs/thesis/lib/python3
.10/multiprocessing/process.py", line 160, in is_alive

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child
process'assert self._parent_pid == os.getpid(), 'can only test a child process'

```

```

AssertionErrorAssertionError: : can only test a child processcan only test a
child process
    assert self._parent_pid == os.getpid(), 'can only test a child
process'assert self._parent_pid == os.getpid(), 'can only test a child process'

```

```

AssertionErrorAssertionError: : can only test a child processcan only test a
child process

```

```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: can only test a child process

: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

Traceback (most recent call last):

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: assert self._parent_pid == os.getpid(), 'can only test a child

```
process'
AssertionError: can only test a child process
can only test a child process

[9] loss 0.0016 | train_acc 1.000 | val loss 0.0643 | val acc 0.994
Epoch 10: 0%|          | 0/2191 [00:00<?, ?it/s]
[10] loss 0.0013 | train_acc 1.000 | val loss 0.0375 | val acc 0.996
Epoch 11: 0%|          | 0/2191 [00:00<?, ?it/s]
[11] loss 0.0013 | train_acc 1.000 | val loss 0.0349 | val acc 0.994
Epoch 12: 0%|          | 0/2191 [00:00<?, ?it/s]
No val-loss improvement for 10 epochs → stopping early.
```

```
[ ]: <All keys matched successfully>
```

1.2.5 Evaluation

```
[ ]: from sklearn.metrics import classification_report, confusion_matrix

model.eval()
all_preds, all_true = [], []

for x, y in test_dl:
    logits = model(x.to(device))
    preds = logits.argmax(1).cpu()
    all_preds.extend(preds.numpy())
    all_true.extend(y.numpy())

test_acc = (np.array(all_preds) == np.array(all_true)).mean()
print(f" Test accuracy: {test_acc:.3f}")

print("\nClassification report (EMBRYO = 1, NO_EMBRYO = 0):")
print(classification_report(all_true, all_preds, target_names=["NO_EMBRYO",
↪ "EMBRYO"], digits=3))
```

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
```

```
Traceback (most recent call last):
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
```

```
Traceback (most recent call last):
```

```
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
```

```

    if w.is_alive():
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
self._shutdown_workers()
        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
            if w.is_alive():
                File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
                    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
                    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():
            File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
                assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError:
            File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
                assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process File "/home/khasion/miniconda3/en
vs/thesis/lib/python3.10/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-

```



```

packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child
process'self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process

```

```

AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'if
w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError:
AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
Exception ignored in:   File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
Traceback (most recent call last):
Exception ignored in:   File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()self._shutdown_workers()

```

```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()self._shutdown_workers()

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
        if w.is_alive():if w.is_alive():

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
assert self._parent_pid == os.getpid(), 'can only test a child process'
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
assert self._parent_pid == os.getpid(), 'can only test a child process'

AssertionError: can only test a child process
AssertionError:
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in:         if w.is_alive():

```

```

        self._shutdown_workers()
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in:      if w.is_alive():
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child
process'<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multip
rocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child
process'<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

AssertionError: can only test a child process
Traceback (most recent call last):
: can only test a child process
Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()Exception ignored in: File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
        self._shutdown_workers()Exception ignored in:
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>

    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
        if w.is_alive():Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive
            if w.is_alive():Traceback (most recent call last):
    File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

        File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive
            self._shutdown_workers()
assert self._parent_pid == os.getpid(), 'can only test a child process' File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

```

```

AssertionErrorself._shutdown_workers()
assert self._parent_pid == os.getpid(), 'can only test a child process' File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

AssertionError    if w.is_alive():
:   File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/pr
ocess.py", line 160, in is_alive
can only test a child process
    if w.is_alive():
:   File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/pr
ocess.py", line 160, in is_alive
can only test a child process
assert self._parent_pid == os.getpid(), 'can only test a child process'
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError    assert self._parent_pid == os.getpid(), 'can only test a child
process'
AssertionError: can only test a child process
: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>

```

```

Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child process
AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'

```

```

AssertionError: AssertionError: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>Exception ignored in:
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>can only test
a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>Exception ignored in:
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
self._shutdown_workers()
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
self._shutdown_workers()      self._shutdown_workers()

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    self._shutdown_workers()

  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers

if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proces
s.py", line 160, in is_alive

if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proces

```

```
s.py", line 160, in is_alive
```

```
assert self._parent_pid == os.getpid(), 'can only test a child
process'AssertionError
: AssertionErrorcan only test a child processassert self._parent_pid ==
os.getpid(), 'can only test a child process'AssertionError
: AssertionErrorcan only test a child process
:
: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():Exception ignored in:
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>    if
w.is_alive():Exception ignored in:
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>    assert
self._parent_pid == os.getpid(), 'can only test a child process'

    assert self._parent_pid == os.getpid(), 'can only test a child process'

Traceback (most recent call last):
AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
: Traceback (most recent call last):
AssertionError File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
: can only test a child process
self._shutdown_workers()can only test a child process
self._shutdown_workers()
```



```

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in:
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
assert self._parent_pid == os.getpid(), 'can only test a child process'
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError
: can only test a child process
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

AssertionError
: can only test a child process
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

self._shutdown_workers()
Exception ignored in: self._shutdown_workers()
Exception ignored in: File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
if w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive
self._shutdown_workers()
Traceback (most recent call last):
if w.is_alive(): File
"/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/
process.py", line 160, in is_alive

```

```

    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
assert self._parent_pid == os.getpid(), 'can only test a child process'
    if w.is_alive():AssertionError:

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
assert self._parent_pid == os.getpid(), 'can only test a child process'
    if w.is_alive():AssertionError:
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
can only test a child process
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
can only test a child process
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionErrorassert self._parent_pid == os.getpid(), 'can only test a child
process'
AssertionError: can only test a child process
: can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():    self._shutdown_workers()
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: AssertionError: can only test a child process
can only test a child process
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at

```

```

0x7a7856b023b0>
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child processAssertionError: can only test a
child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at
0x7a7856b023b0>Exception ignored in: <function
_MultiProcessingDataLoaderIter.__del__ at 0x7a7856b023b0>
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
Traceback (most recent call last):
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1604, in __del__
    self._shutdown_workers()
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/site-
packages/torch/utils/data/dataloader.py", line 1587, in _shutdown_workers
    if w.is_alive():
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
  File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
AssertionError: can only test a child processAssertionError: can only test a
child process

File "/home/khasion/miniconda3/envs/thesis/lib/python3.10/multiprocessing/proc
ess.py", line 160, in is_alive

```

```

    assert self._parent_pid == os.getpid(), 'can only test a child process'
    assert self._parent_pid == os.getpid(), 'can only test a child process'
    AssertionError: can only test a child process

```

```

AssertionError: can only test a child process

```

```

    Test accuracy: 0.994

```

```

Classification report (EMBRYO = 1, NO_EMBRYO = 0):

```

	precision	recall	f1-score	support
NO_EMBRYO	1.000	0.970	0.985	9198
EMBRYO	0.992	1.000	0.996	34188
accuracy			0.994	43386
macro avg	0.996	0.985	0.990	43386
weighted avg	0.994	0.994	0.994	43386

1.2.6 Save Model + quick demo

```

[ ]: model.eval()
    # Create output directory if needed
    import os
    os.makedirs("embryo_presence_models", exist_ok=True)

    MODEL_PATH = "embryo_presence_models/embryo_presence_resnet18_all_focal_planes.
    ↪pth"
    torch.save(model, MODEL_PATH)
    print(f"Model trained on all {len(FOCAL_PLANES)} focal planes: {FOCAL_PLANES}")
    print(f"Model saved to {MODEL_PATH}")

    # Visualise a few predictions
    idx_present = [i for i, (_, lbl, _) in enumerate(test_ds.samples) if lbl == 1]
    idx_absent = [i for i, (_, lbl, _) in enumerate(test_ds.samples) if lbl == 0]

    n_each = 10
    sel_inds = random.sample(idx_present, n_each) + random.sample(idx_absent,
    ↪n_each)
    random.shuffle(sel_inds)

    images = torch.stack([test_ds[i][0] for i in sel_inds])
    labels = torch.tensor([test_ds[i][1] for i in sel_inds])

    preds = model(images.to(device)).argmax(1).cpu()

    fig, axes = plt.subplots(4, 5, figsize=(12, 6))
    for ax, img, lab, pred in zip(axes.flatten(), images, labels, preds):

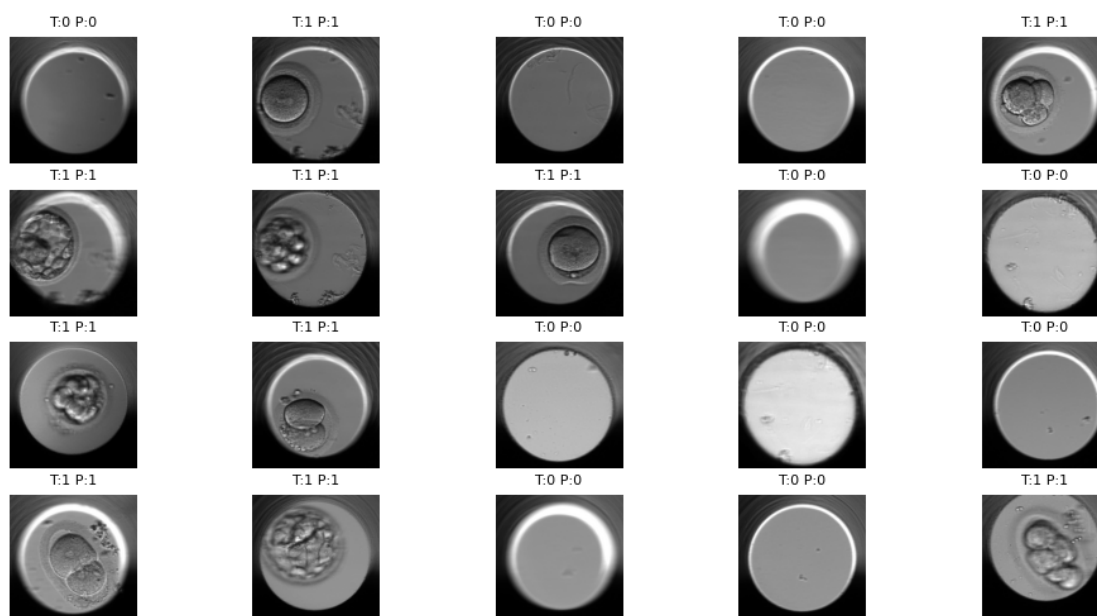
```

```
ax.imshow(img.squeeze(0).numpy()*0.5 + 0.5, cmap="gray")
ax.set_title(f"T:{lab.item()} P:{pred.item()}", fontsize=8)
ax.axis("off")
```

Model trained on all 7 focal planes: ['F0', 'F-15', 'F-30', 'F-45', 'F15', 'F30', 'F45']

Model saved to

embryo_presence_models/embryo_presence_resnet18_all_focal_planes.pth



1.2.7 Grad-CAM montage: 50% Embryo, 50% No Embryo (test frames)

```
[ ]: # -----
# Grad-CAM montage: 50 % EMBRYO, 50 % NO_EMBRYO (test frames)
# -----
from torchcam.methods import GradCAM
import torch.nn.functional as F

model.eval() # keep weights frozen
cam_extractor = GradCAM(model, target_layer=model.layer4[-1])

# ----- 1. Balanced sampling -----
random.seed(0)

idx_embryo = [i for i, (_, lbl, _) in enumerate(test_ds.samples) if lbl == 1]
idx_empty = [i for i, (_, lbl, _) in enumerate(test_ds.samples) if lbl == 0]

n_each = 6
```

```

sel_inds = random.sample(idx_embryo, n_each) + random.sample(idx_empty, n_each)
random.shuffle(sel_inds)

# ----- 2. Compute CAMs -----
images, labels, cams = [], [], []
for idx in sel_inds:
    x, lbl = test_ds[idx]          # x: [C,H,W] tensor
    x_batch = x.unsqueeze(0).to(device)

    # Forward *with* gradients
    logits = model(x_batch)

    # CAM for the predicted class (or use lbl to see GT class)
    cls_id = logits.argmax(1).item()
    cam_small = cam_extractor(cls_id, scores=logits)[0]
    cam_small = cam_small.squeeze(0)

    # Upsample to input resolution
    cam_up = F.interpolate(cam_small.unsqueeze(0).unsqueeze(0),
                           size=(IMG_SIZE, IMG_SIZE),
                           mode='bilinear',
                           align_corners=False).squeeze()

    cam_up = (cam_up - cam_up.min()) / (cam_up.max() - cam_up.min() + 1e-6)

    images.append(x)
    labels.append(lbl)
    cams.append(cam_up.cpu())

# ----- 3. Plot grid -----
n_cols = 6
n_rows = math.ceil(len(images) / n_cols)
fig, axes = plt.subplots(n_rows, n_cols, figsize=(14, 2.5*n_rows))

for ax, img, lbl, cam in zip(axes.flatten(), images, labels, cams):
    base = img.squeeze()*0.5 + 0.5      # un-normalise to [0,1]
    ax.imshow(base.numpy(), cmap='gray')
    ax.imshow(cam.numpy(), cmap='jet', alpha=0.4)
    ax.set_title("EMBRYO" if lbl==1 else "NO_EMBRYO", fontsize=8,
                 color='lime' if lbl==1 else 'red')
    ax.axis('off')

plt.tight_layout()
plt.show()
# -----

```

